This thesis extends a genetic-programming hyper-heuristic (GPHH) baseline for the strongly NP-hard multi-mode resource-constrained project scheduling problem. A staged methodology first screens a wide field of candidate modifications cheaply, then evaluates the survivors at full statistical power on MMLIB50 and MMLIB100. Of the screened variants, including a diagnosis-driven graft-based variation scheme, only one, epsilon-lexicase selection, together with two independently motivated extensions, critical-path local search and non-renewable (NR) resource terminals, is carried forward. NR terminals, which expose previously invisible non-renewable resource state to the evolving rule, deliver a large, statistically significant improvement in schedule quality that replicates robustly across benchmark scale and on the complete held-out test set; the accompanying feasibility gain proves largely confined to instance classes seen during training. Lexicase selection reliably improves training fitness but this advantage does not transfer reliably to held-out test fitness, and the gap widens rather than narrows at larger scale. Critical-path local search improves schedule quality on the smaller benchmark, but only under one of the two schedule generation schemes, and the improvement does not survive the move to the larger one.